SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I H. Divyaa Dharshini 192420174 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. Divyaadharshini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	1	2	1
Organization	15%	1	1	1	2
Accuracy	10%	1	1	1	0
Clarity	5%	0	1	0	1

Total Marks **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-struct ured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization , incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CSA0410 : COMPUTER NETWORKS

H. Divyaa Dharshini
192426014
CO1 - AT1

8) A packet experiences the following processing delays at different OSI layers: Application layer = 4ms, Transport layer = 3ms, Network layer = 5ms, Data link layer = 2ms, and Physical layer = 1ms. The propagation delay is 18 ms, transmission delay is 12ms. Calculate the total end to end delay experienced by the packet and explain how each layer contributes to communication process.

Introduction:

End-to-end delay is the total time taken by a packet to travel from a sender to the reciever through a network

→ When data travels across a network, it experiences different types of delays. These include processing delay at different OSI layers, transmission delay and propagation delay.

→ The total end-to-end delay is calculated by adding all the given delays.

Formula:

Total delay = Processing delay + Transmission delay + propagation delay.

Processing delay is the sum of the delays given for all five OSI layers.

Given data:

LAYER (or) TYPE OF DELAY	TIME
Application layer	4 ms
Transport layer	3ms
Network layer	5ms
Data link layer	2ms
Physical layer	1 ms
Propagation layer	18ms
Transmission layer	12ms

Step by step calculation:

→ Step 1: Calculate the total processing delay.

$$D_p = 4 + 3 + 5 + 2 + 1$$

$$= 15ms$$

Total processing delay = 15ms

Step 2:
Identify the other delays

→ Transmission delay = 12 ms

→ Propagation delay = 18 ms

Step 3: Calculate total end to end delay

$$D_{Total} = 15 + 12 + 18$$
$$= 45 ms$$

FINAL ANSWER

45 milliseconds

OSI layer contribution to communication:

→ Application layer

→ Transport layer

→ Network layer

→ Data link layer

→ Physical layer

Conclusion:

The packet experiences a total processing delay of 15 ms across the five OSI layers, a transmission

delay of 12 ms, and a propagation delay of 18 ms. Adding these values gives a total end-to-end delay of 45 ms.

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b) Before transmission, the presentation layer compresses a 900MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size and total percentage reduction. Explain how the presentation layer improves transmission efficiency.

Given:

Original file size = 900 MB

Compression = 40%.

Encryption overhead = 5%.

Calculate the compressed file size:

$$\text{compressed size} = \text{original size} \times \frac{(100-40)}{100}$$

$$= 900 \times \frac{60}{100}$$

$$= 540 \text{ MB}$$

Calculate the encrypted file size:

$$\text{Encrypted size} = 540 + \left[540 \times \frac{5}{100}\right]$$

$$= 567 \text{ MB}$$

Calculate the total percentage reduction

Reduction = Original - encrypted size

= 900 - 567

= 333 MB

Percentage reduction = (Reduction / Original size)

$= \frac{333}{900} \times 100$

= 37 %.

Total percentage reduction = 37 %.

→ The presentation layer basically improves the transmission efficiency.

→ It is done by compressing data and by reducing amount of the data being transferred over the network.

→ Here it also supports to encryption to protect the sensitive information from unauthorized data.

→ Therefore, it helps improve bandwidth utilisation and data security.

4) An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The application layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how application layer provides reliable services to users.

Given:

Total data = 3 GB

Throughput = 120 Mbps

Request size = 3 MB

Here 1 GB = 1024 MB

→ Calculate the total no. of requests

Total data = 3 × 1024

= 3072 MB

No. of requests = $\frac{\text{Total data}}{\text{Request size}}$

= 3072 / 3

= 1024 requests.

→ Calculate transmission time

1 MB = 8 megabites.

1) Calculate data per frame

Useful data per frame = Frame size - Overhead

= 1600 - 80

= 1520 bytes.

2) Calculate the total number of frames

Assume 1 MB = 1,000,000 bytes

Total useful data = 80 × 1,000,000

= 80,000,000 bytes

Number of frames = $\frac{\text{Total useful data}}{\text{Useful data per frame}}$

= $\frac{80000000}{1520}$ = 52,631.58

Total frames = 52,632

3) Calculate total overhead transmitted

Total overhead = Number of frames × Overhead per frame

= 52,632 × 80

= 4,210,560 bytes

≈ 4.21 MB

Total data = 3072 × 8

= 24,576 megabits

Transmission time = Total data / Throughput

$$= \frac{24576}{120} = 204.8 \text{ s}$$

→ The application layer provides network services to user application through protocol such as FTP.

→ FTP supports file transfer between computers.

→ It allows user to upload and download files, where error handling and reliable delivery depend on the underlying transport protocol, typically TCP.

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9) Given:

→ Useful data = 80 MB

→ Frame size = 1600 bytes

→ Protocol overhead per frame = 80 bytes.

calculate transmission efficiency

Transmission efficiency = $\frac{\text{useful data}}{\text{total transmitted data}} \times 100$

$= 80,000,000 + 4,210,560$

$= 84,210,560$ bytes

EFFICIENCY = $\frac{80000000}{84210560} \times 100$

$\approx 95\%$

FINAL ANSWER

Total frames = 52,632

Total overhead $\approx$ 4.21 MB

Transmission efficiency $\approx$ 95%